

Controlling the synchronization and symmetry breaking of coupled bacterial pili on active biofilm carpets

Reviewed Preprint

v1 • September 29, 2025

Not revised

Baha Altın, Yusuf Ilker Yaman, Enes Talha Günay, Alp Ünlü, Yiğithan Gediz, Neslihan Gedik, Bora Karadaş, Mustafa Başaran, Coşkun Kocabaş, Şahin Özdemir, Aşkın Kocabaş 

Department of Physics, Koç University, Istanbul, Türkiye • Sciences and Engineering Program, Harvard University, Cambridge, United States • Department of Physics, Undergraduate Program, Boğaziçi University, Istanbul, Türkiye • Department of Materials, University of Manchester, Manchester, United Kingdom • Department of Engineering Science and Mechanics, The Pennsylvania State University, University Park, United States • Department of Electrical and Computer Engineering, Saint Louis University, St Louis, United States • Koç University Surface Science and Technology Center, Koç University, Istanbul, Türkiye

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **important** study concerns the propagation of waves in bacterial biofilms, bridging active matter physics and bacterial biophysics. While the experimental observations are **solid**, the theoretical interpretation and model validation are currently **incomplete** and require further refinement. This work will be of interest to microbiologists, biophysicists, and researchers studying collective behavior in biological systems.

<https://doi.org/10.7554/eLife.107609.1.sa3>

Abstract

In the low Reynolds number regime, active biological systems utilize nonreciprocal cyclic activities to achieve motility, as seen in the spinning of bacterial flagella and the beating of cilia. Coupling among these active mechanical components leads to synchronization, and emergence of metachronal waves. Here, we report that biofilms of *Pseudomonas nitroreducens* form active carpets surfaces textured with various topological defects generating large-scale propagating waves. On these active structures, nonreciprocally coupled extension and retraction activities of bacterial pili drives the collective oscillations. Surprisingly, this collective behavior exhibits broken left-right symmetry. We discover that this symmetry-breaking process is primarily driven by an aging-related frequency gradient across the biofilm. Leveraging these insights, we further demonstrate the ability to control the collective dynamics of these waves, including symmetry breaking, transitions from spiral waves into target and propagating plane waves by manipulating the biofilm properties. Overall, our findings illuminate the fundamental role of nonreciprocally interacting active components in regulating synchronization, collective dynamics, and symmetry-breaking phenomena in biological systems.

Introduction

At low Reynolds numbers regime, overcoming reversibility constrain to gain motility, is a challenging task which requires innovative solutions. Microorganisms utilize a variety of specialized active mechanical components to solve these difficulties by performing nonreciprocal motion^{1–4}. Typical examples include the rotating helical flagella and beating elastic cilia. These active mechanical systems have the ability to execute cyclic motions, thereby breaking time-reversibility constraints⁵. In addition, due to the coupling among these active microstructures, complex synchronization phenomena also known as metachronal waves could emerge^{6–8}. From these perspective coupling, synchronization and collective behaviors are intrinsically related and ubiquitous in nature. Such widespread dynamical processes are also biologically essential, as seen in cilia-driven mucus transport in lungs⁹, nodal flow in developing embryos¹⁰, central motor generators¹¹ and spinal cord development¹². These intricate synchronization behaviors share the same underlying dynamical principles that surprisingly align with non-Hermitian physics of active systems^{13–16}. All these active systems are capable of injecting and transferring energy across the system by breaking certain symmetries.

Recent advances, especially those in the field of active matter physics^{17–22}, have illuminated critical non-Hermitian features, such as chiral states where PT symmetry is broken. In this broken mode system could sustain the phase difference between fundamental fields or the modes which further trigger the directional energy transfer between the active compartments and generate travelling waves. Notably this non-Hermitian process is particularly driven by nonreciprocity in the coupled mechanical system^{8,17,23}. Although non-Hermiticity has its roots in quantum mechanics^{14,24} and the fundamental principles have been broadly studied in photonics²⁵, electronics, acoustics, optomechanics^{26,27}, superconducting qubits²⁸, trapped ions^{29,30}, single-spin systems³¹, and in light-matter interactions³². This process is often achieved not only using strong nonlinearities, but also utilizing nonreciprocal interactions and coupling^{33–35}. On the other hand, nonreciprocity is a very common feature in biological systems. Such as hydrodynamic interactions³⁶ or prey predator³⁷ relations could easily break action reaction symmetry and results in exotic collective behaviors. Extending the basic concepts from non-Hermitian physics to biological domain holds promise for shedding light on the fundamental principles of symmetry breaking in active and, more crucially, in complex biological systems.

Various biological models with specific active micromechanical components have been studied. The most notable biological platforms are ciliated epithelial cell³⁸, bacterial carpet^{39,40}, starfish embryo⁴¹, social amoeba^{42–44} and walking placozoa⁴⁵. Addition to flagellum or cilia, bacteria could also use nano filaments known as type IV pili to gain motility on surfaces where the rotating flagella do not work effectively. Recent studies also highlighted the importance of pili controlling the collective behaviors of bacteria^{46–49} and also large-scale oscillations in the form of biofilm.

During our recent experiments, we noted striking dynamic patterns and unidirectionally propagating waves on the surface of *Pseudomonas Nitroreducens* (PN) bacterial biofilms. These unexpected collective activities ranging from spiral to planar wave formations, interestingly aligned with recent predictions of active matter physics^{23,50}. These activities are reminiscent of metachronal waves observed in various biological systems. We also observed that these collective activities are driven by the coupled pili activity of the bacteria. Recently, large-scale pili-driven activities and spiral waves were also identified in *Pseudomonas aeruginosa*⁴⁹. The main distinction of our observed metachronal wave lies in its localized behavior on the surface and broken left-right symmetry across the biofilm, which we refer to as the active biofilm carpet. In

this study, we aim to provide a comprehensive experimental approach including biological, physical and theoretical tools to be able to understand and control these collective behaviors and also the symmetry breaking process of these biological systems.

Bacterial biofilms develop into multicellular communities characterized by interdependent biological structures. Understanding the dynamics of these dense biological formations is vital, given that their collective actions can amplify their pathogenic potential. The collective movement of pili and its variability among strains offers a crucial foundation for understanding the influence of pili activities on pathogenicity. Moreover, pili is also a primary target protein for next-generation medications and antibacterial therapies. In essence, comprehending the effect of pili dynamics on the collective behavior of biofilms can greatly influence human health by preventing severe infections.

Results

Emergence of metachronal waves on active biofilm carpets

Bacterial biofilms are commonly used in various experiments particularly to investigate host-pathogen interactions. During our recent screen of biofilm library for nematode *C.elegans*⁵¹, we observed that *Pseudomonas Nitroreducens* (PN, Materials and Methods) bacterial biofilms exhibit rhythmic activities approximately 10 hours post-inoculation (**Figure1a-b**). In their firing phase, faint propagating pairs of spiral wave patterns emerged on the biofilm's surface. These waves have a periodicity of around 2-7 minutes (**Figure1c**). We also determined that the visibility of these waves improves dramatically with oblique contrast and polarized light-based microscopy (**Figure1a-b**, Supplementary Video1-2). Later on, these waves were very dynamic and converged to various combinations of spiral, target and planar waves. The most striking feature of these waves is their broken left-right symmetries. They, often unidirectionally propagate from right to left (edge to the center) or radially in inward direction across the colony (**Figure1a-b**). Furthermore, the contrast of the wave's switches between the dark to bright sharp lines depending on the direction of propagation. This suggest that the front kink of these waves scatters the light asymmetrically (**Supplementary Figure 1**). We observed that these waves predominantly propagate on the biofilm's surface rather than its bulk region (**Figure1e**). This critical feature is particularly revealed by the nature of our optical imaging systems which is very sensitive to the surface topography. To validate this observation, we employed cytoplasmic GFP for imaging. Yet, no waves were detected in the green fluorescence channel (**Supplementary Figure 2**). Similarly, when we labeled extracellular DNA with a dye, the waves remained unobserved in the fluorescence signal (**Supplementary Figure 2**). Moreover, close-up imaging (100X) highlighted the bacterial displacement together with clear optical dark or bright contrast on the surface (Supplementary Video3). Imaging growing biofilm starting from a single bacterium clarifies that oscillations first starts from dense regions and gradually propagate through the entire colony (Supplementary Video4). Further motivated by recent studies^{48,49}, we also tested the *Pseudomonas aeruginosa* strain PA14 under the same conditions. Similarly, PA14 exhibited weak but comparable wave patterns, though within a shorter time window (**Supplementary Figure 3**).

To delve deeper into these phenomena, our next focus was on the necessary media, given that the original experiments were conducted on *C. elegans* using Nematode Growth Media (NGM) plates (see Materials and Methods). Interestingly, we found that these activities were exclusive to NGM. Substituting NGM with a standard LB plate did not reproduce the same dynamics; instead, the bacteria grow rapidly. Yet, upon omitting yeast extract and replacing bactotryptone with bactopectone, a major nutritional component, from the LB media PN resumed its activity

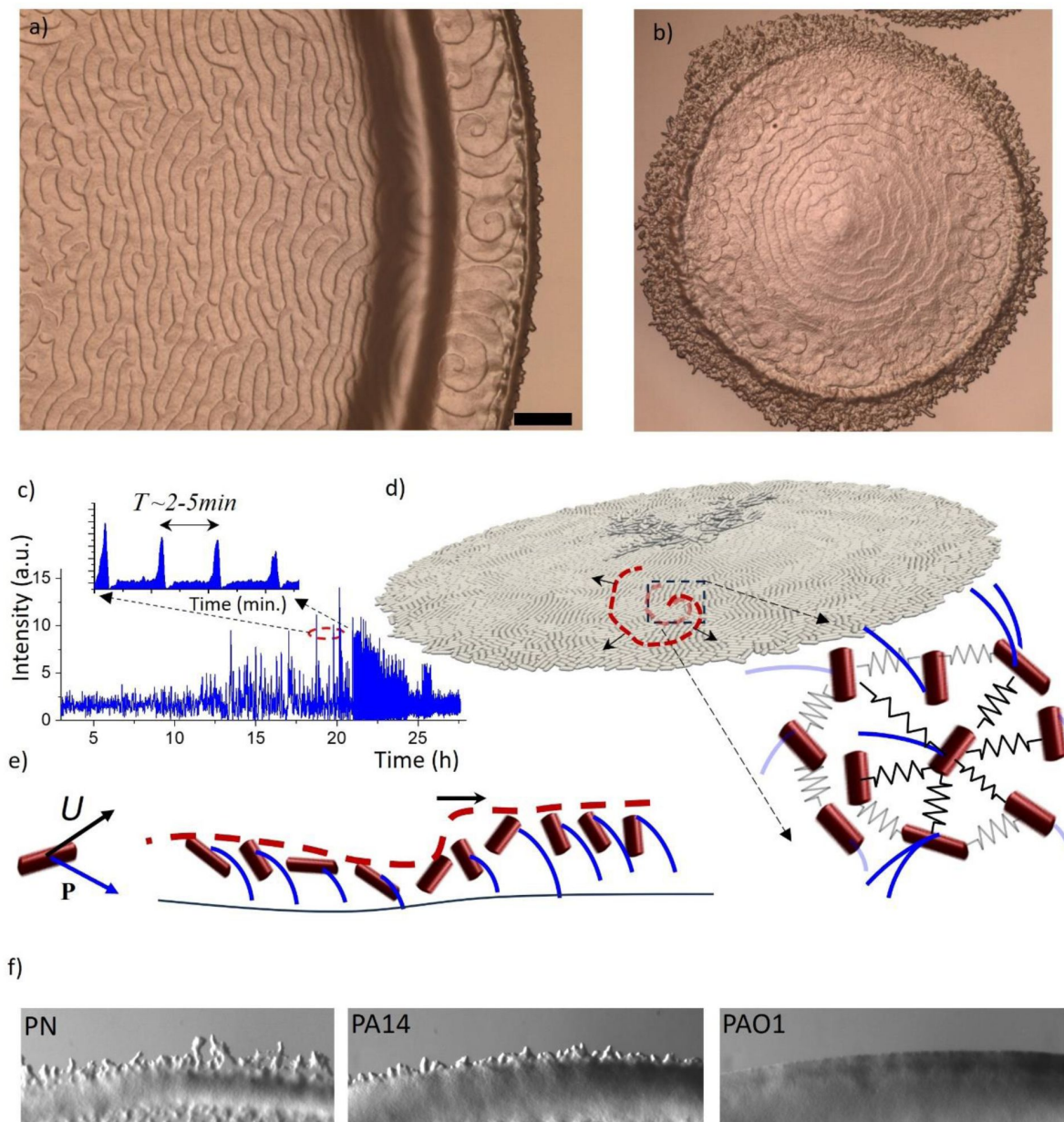


Figure 1.

Emergence of unidirectionally propagating waves on bacterial biofilms.

(a-b) Optical imaging shows spiral, planar (a), and radially shrinking waves (b) propagating unidirectionally on biofilm surfaces. Left-right or radial symmetry is generally broken on growing biofilm surfaces. Scale bar 250 μm . (c) Time response of optical scattering signal indicating the firing state of coupled pili dynamics. (d) Schematic representation of the coupling mechanism of pili on biofilm surfaces, modeled as an active solid. The oscillatory extension and retraction of pili act as active units. (e) Collective behavior of elastically coupled active biofilm surfaces characterized by local displacement (U) and pili polarization (P). Propagating waves remain localized on the surface and travel toward the direction of the sharp rising edge. (f) Optical images of the leading edge of the growing biofilm with fingering instabilities. PN and PA14 show strong fingers compared to PAO1.

(**Supplementary Figure 5** [↗](#)). These findings underscore the significance of nutrient restriction as a primary factor. Remarkably, these observations are unexpected, but become evident due the use of NGM plate, which is not common in microbiology experiments.

Although PA14 generates similar waves we observed that PAO1, a close relative of PA14, failed to produce waves on both NGM and LB plates (**Supplementary Figure 4** [↗](#)). Worth noting, PAO1 is also an important human pathogen causing *Cystic Fibrosis* (CF). Finally, through this comprehensive screening effort, we successfully identified a group of bacteria, offering the potential to examine various factors, from genetics to physical domain, influencing the emergence of these spiral waves. Both PAO1 and PA14 possess robust genetic toolkits, while PN strain display a clear and prolonged active carpet state for detailed physical investigations. The other critical difference between all these biofilms is the finger formation around the leading edge. Unlike PAO1, PN and PA14 form very strong instabilities leading unstable extensions (**Figure1f** [↗](#)).

We then shifted our attention to the genetic distinctions between PN, PA14 and PAO1. Initially, we first explored the contribution of bacterial pili and flagella as an active mechanical structure. We found that PA14:ΔPilB and PA14:ΔPilA mutations eliminated the waves but not PA14:ΔFilK (**Supplementary Figure 3** [↗](#)). This suggests that activity is exclusively originating from pili dynamics and the original dynamics is similar to the recent work⁴⁹[↗](#), but activities are localized to the surface. We then explored the potential of hyper piliation, which might amplify the coupling of pili on the biofilm surfaces. Yet, the PAO1 ΔPilH mutant, known for its polarized and hyper piliation condition^{46,47}[↗](#), did not manifest the waves but provided only small fluctuations on the surface (**Supplementary Figure 4** [↗](#)). Then, we pinpointed that a significant variation was in the pili subunit groups⁵²[↗](#). While PA14 possesses group 3 Pilin, PAO1 has group 2 Pilin. It's essential to highlight that all these bacteria have type IV pili, though they might be categorized into separate groups. Currently, our hypothesis posits that diverse pili types might exhibit varied physical characteristics such as elasticity or rigidity. Our understanding of PN at the genetic level remains limited but it shares strong similarity with PA. Nonetheless, the various genome sequences of PN are accessible. Our analysis of their pili related genomic region revealed a resemblance to the group 4 and 5 pili (**Supplementary Figure 6** [↗](#)). All these firing pili groups 3 and 4, 5 have additional accessory genes in the genome and they may contribute to the expression or folding of these pilins. Together with the unstable edge fingering (**Figure1f** [↗](#)), the comparison of PN and PA strains support the idea that group of pili and particularly pili expression play a critical role in driving waves on these bacterial biofilms.

Modelling of propagating spiral waves

To get more intuition about the dynamics of the waves and emergence of the broken symmetry across the biofilm, we next focused on the mathematical modelling. From an active matter perspective, spiral waves can emerge in non-equilibrium and excitable media^{53–55}[↗](#) as well as in coupled oscillatory systems with delays⁵⁶[↗](#). Despite these systems exhibiting similar wave patterns, the internal mechanisms that trigger the waves vary significantly. Recent studies particularly on the concept of nonreciprocity provides comprehensive approach^{23,50}[↗](#). It is important to note that coupling between active mechanical units could introduce nonreciprocity into the system due to local symmetry-breaking processes. Inspired from the previous studies on the dynamics of cilia^{17,57}[↗](#), and recent robotic system⁵⁸[↗](#) we proposed that coupled active pili system could also show similar limit cycles oscillations defined by bacterial displacement (**U**) and the polarization of the bacterial pili (**P**) (**Figure1e** [↗](#)). To study these phenomena, we first utilized a minimal phenomenological phase field model that includes a nonreciprocal coupling term. The oscillatory dynamics of the model are driven by oscillatory extension and retraction processes. The Kuramoto based model^{23,56,59}[↗](#), which effectively captures the phase dynamics of

coupled oscillators, is extended here by incorporating a delay term (α) that disrupts the odd symmetry—representing action and reaction symmetry between interacting oscillators—and enriches the system’s dynamic repertoire. This modified model is described by the equation:

$$\frac{d\theta_i}{dt} = w_0 - b \sin(\theta_i) - K \sum_j \sin(\theta_i - \theta_j - \alpha)$$

Here, θ represents the local phase of individual pili oscillations (The phase of parameter space defined by **U** and **P**, **Figure 1e**). We assume interactions only among nearest neighbors, where w_0 represents the uniform intrinsic oscillation frequency of an isolated oscillator. The term α specifically break odd interaction symmetry between pili, facilitating the formation of travelling waves. Note that traveling waves indicate this broken PT symmetry between these fields. It’s important to emphasize that the detailed biophysical mechanisms behind this nonreciprocity remain unclear. Several possibilities, such as force relaxation dynamics or hydrodynamics interactions, could also result in similar nonreciprocal behavior. Our phenomenological model accurately captures the principal features of these phenomena. While reciprocally coupled oscillators ($\alpha = 0$) tend to achieve a globally synchronized state (**Figure 2a**), nonreciprocity favors the formation of traveling waves (**Figure 2b-c**, Supplementary Video5). The excitability term (**b**) shapes the pulsative nature of the waves (illustrated in **Figure 2d-e**) by amplifying the effects of specific phases within the coupling mechanism.

$$\begin{aligned} \partial_t \mathbf{U} &= \pi \mathbf{P} + \mathbf{F}_e \\ \partial_t \mathbf{P} &= (\mathbf{P} \times \partial_t \mathbf{U}) \times \mathbf{P} + \frac{1 - \mathbf{P}^2}{2} \partial_t \mathbf{U} - D_r \mathbf{P} \end{aligned}$$

Next, we turned our focus to continuum simulations to capture the essential properties of these dynamic processes (Supplementary Video6). The recently developed active solid model provides a broad and robust framework for studying collective behaviors of interacting active components, spanning from robotic systems to human crowds. The core idea of this model is based on two nonreciprocally coupled fields: displacement (**U**), representing bacterial displacement within the biofilm lattice, and polarization (**P**), representing the orientation of pili, which exert active forces whose direction can be modulated by the displacement field (**Figure1d**). The dynamics of the active biofilm lattice can be described by coupled equations, where $\mathbf{F}_e$ denotes elastic force and D_r represents polarization relaxation. The underlying rationale of these equations is that retracted pili exert active forces on bacteria, which are elastically coupled to the biofilm structure. Local biofilm displacement can guide the growth direction of pili extension, influenced by local deformation. Importantly, since pili are anchored to the substrate, their orientation dynamically adjusts based on extension processes. Our simulations demonstrate that the active solid model successfully captures critical dynamical phenomena, such as the formation of spiral and planar waves (**Figure 2 f-g**). As anticipated, fields **U** and **P** develop a characteristic phase difference, indicative of limit cycle oscillations. Moreover, analysis of vector configurations around defect cores reveals directional reversals of displacement **U** on opposite sides of defects, highlighting the fundamental mechanism underlying defect formation, where **U** diminishes, and the phase (θ) becomes undefined.

Controlling the transition between the waveforms on active biofilm carpet

Extensive parameter testing has revealed that our models not only capture the emergence of spiral waves but also predicts transitions into target and plane wave solutions which we commonly observe in grooving biofilms. During the transition to target waves, two oppositely spinning spiral waves with topological charges of +1 and -1 merge, forming a symmetrically expanding target wave. Subsequently, fast oscillating plane waves progressively dominate the entire simulation domain (**Figure3 a-c**, Supplementary Video5).

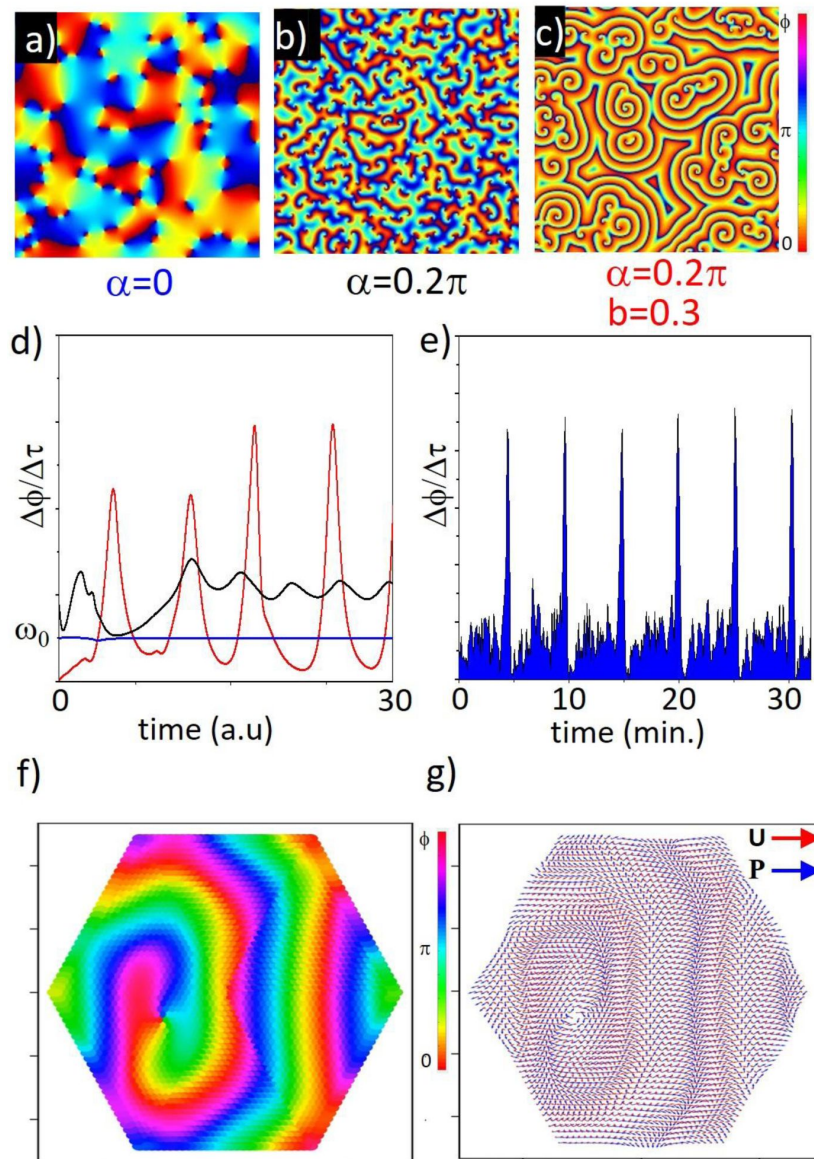


Figure 2.

Numerical modeling of coupled pili dynamics as an active carpet.

(a-c) Numerical simulation results based on the nonreciprocal Kuramoto phase-field model. Nonreciprocal coupling term (α) among bacteria drives the emergence of spiral waves, while the excitability term (b) of the mechanical system induces pulsatile behavior. (d) Numerical simulations of local oscillation frequencies under varying conditions ($\alpha = 0$, $\alpha = 0.2\pi$, and $b = 0.1\pi$). (e) Experimental measurements of pulses on biofilm surfaces confirming pulsatile responses. (f-g) Numerical simulations of elastically coupled biofilm structures using the active solid model. Active solids exhibit large-scale spiral and planar wave formations. Displacement (U) and pili polarization (P) fields highlight the essential phase difference necessary for wave propagation and limit-cycle oscillations.

Motivated by our simulation results, we hypothesize that our active carpet system could also facilitate similar transitions between different forms of traveling waves^{64,65}, offering a means to externally control the dynamics of the system and may explain the details of broken symmetries. We observed that adding a water droplet to the biofilm surface led to the transformation of spiral waves into target and planar waves (**Figure 3d-f**). To recover the spiral waves, we increased the surface temperature by gently heating the system. As expected, with the temperature rise, the leading edge of the planar waves first became noisy, and multiple target waves randomly appeared then finally converge to spiral waves (Supplementary Video7). These findings are in interesting similarity with recent theoretical predictions in different nonreciprocal systems^{50,64}.

Additionally, by incorporating the water-soluble polymer polyethylene glycol (PEG) into the biofilm, we managed to not only alter wave propagation patterns but also direct the movement of inward growing target waves. Intriguingly, the addition of small drop of PEG modified the intrinsic oscillation period of the pili dynamics, creating a radial gradient profile largely due to the slow evaporation rate and uneven deposition of PEG on the surface. This inward wave guidance is particularly controlled by frequency gradient (**Figure 4a-d**). The center has slow oscillations than the edge. This gradient profile is the critical feature to understand the basics of broken symmetry observed in the circular biofilms. In naturally growing colony oscillation frequency is also varying (**Figure 4a-b**). We hypothesize that this variation is age dependent where the center of the colony is slowing down due to the nutrition or oxygen depletion process. Before delving deeper into this interesting symmetry-breaking process, we investigated how the aging of the biofilm affected the oscillation dynamics. This is because symmetry breaking occurs at late stage where the aging is critical. To do so, we first measured the oscillation periods as the uniform colony aged. We observed that the periods of these spiral waves continuously increased (**Figure 5a-b**). These results suggest that the intrinsic oscillation dynamics of the Pili decreased, likely due to depleted nutrients or oxygen, during first 24 hours.

Further, we examined the growing biofilm starting from a thin inoculation strip, we reproducibly achieved a broken left-right symmetry for plane waves across the colony (Supplementary Video8). The waves propagated from the edges toward the center. This asymmetry became particularly pronounced a few days after inoculation, during which the colony edges continued to grow while the center remained stationary (**Figure 6a, b**). To finally confirm this observation, we numerically solve the coupled oscillators under varying frequency. Similarly, we found similar broken left-right symmetry where the planar waves propagating towards the slower region of the biofilm center (**Figure 6c, d**, Supplementary Video9).

Controlling left-right symmetry breaking

As a final step we focused on how to control the symmetry breaking and dynamics of metachronal waves. We noticed that temperature adjustments also reversibly effect the oscillatory behavior of the waves which increased the oscillation frequency (**Figure 7a, b**). This suggest that temperature could be also used as a complementary technique to dynamically guide the wave propagation by controlling the frequency gradients. To do so we embedded metallic pipes in the bulk agar plate and tuned the local temperatures (Supplementary Figure 7). Setting different temperatures in these pipes creates temperature gradient ($\nabla T \sim 10^\circ\text{C}/\text{cm}$). This gradient profile converted spiral waves quickly to planar waves, propagating from warmer (31°C) to cooler (21°C) areas. Similarly, when we switched the temperature profiles wave propagation also reversed (**Figure 7c-d**). This result indicates that spatial temperature variations can effectively dictate wave propagation directions. These comprehensive observations confirm that the developed frequency gradients of intrinsic Pili oscillations is the critical feature breaking symmetry and controlling the propagation directions of the waves.

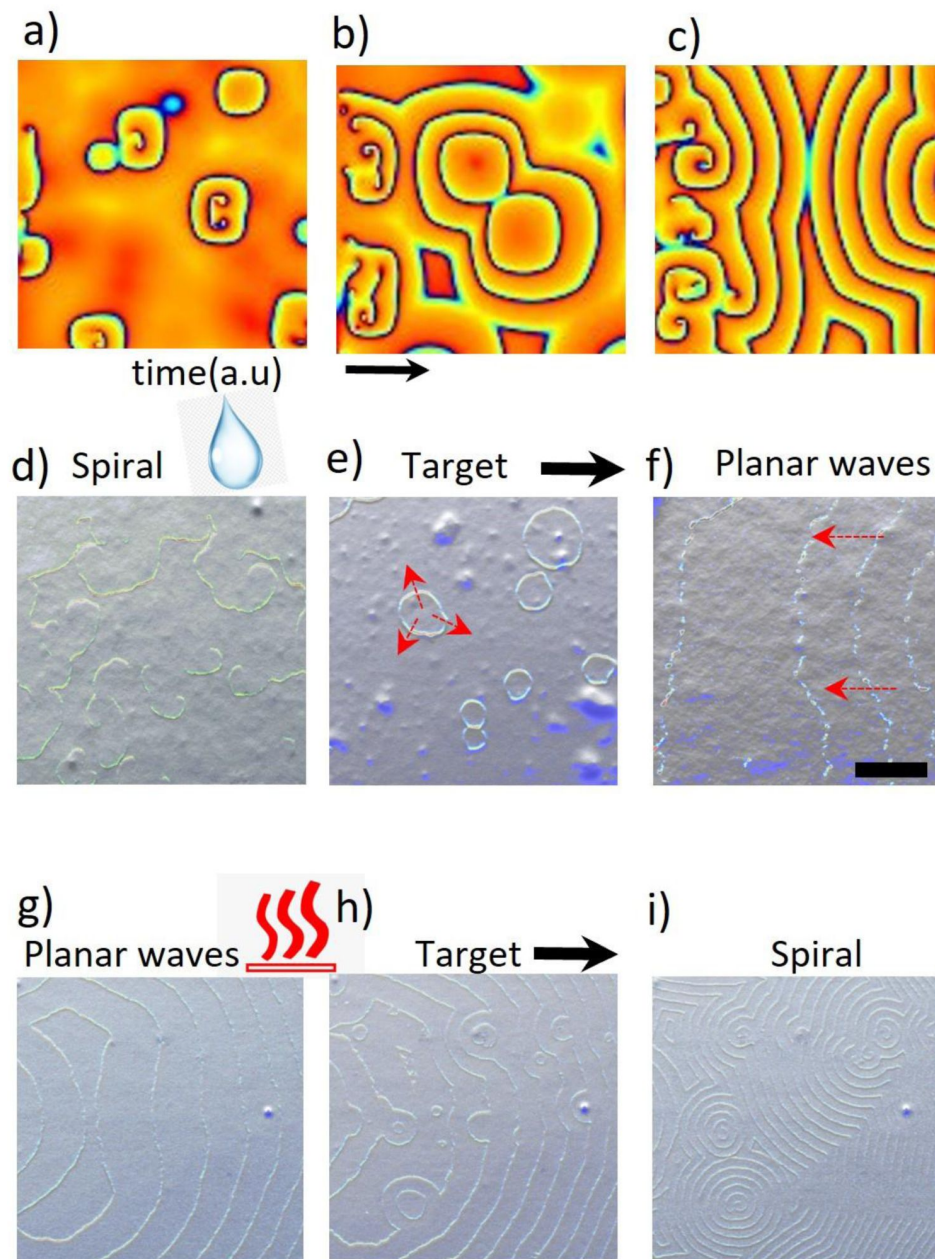


Figure 3.

Controlling transitions between spiral, target, and planar waves.

(a-c) Numerical simulations illustrating transitions from spiral waves to target and planar waves. Pairs of spiral waves merge, forming topologically neutral target waves, which eventually give rise to planar waves dominating the biofilm surface. (d-f) Optical imaging of biofilm surfaces demonstrating similar transitions between spiral, target, and planar waves experimentally triggered by adding a water droplet. Red arrows indicate wave propagation direction. (g-i) Controlled recovery of spiral waves achieved by heating biofilm surfaces, removing excess moisture, and facilitating re-emergence of spiral waves.

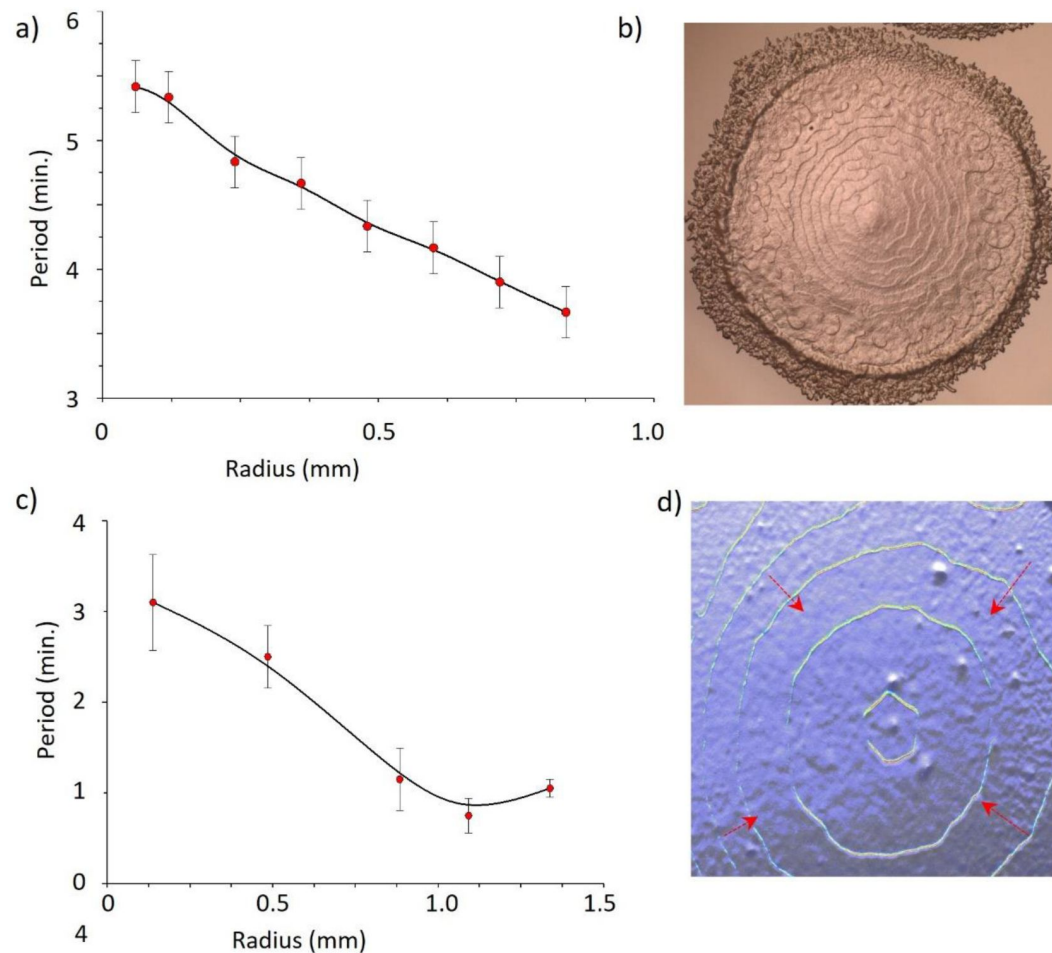


Figure 4.

Controlling dynamics of inward propagating waves.

(a) Optical imaging of inward propagating waves within a circular biofilm structure. (b) Period of pili oscillations decreases toward the colony center. (c) Application of a small droplet containing PEG creates a radially varying period profile, guiding inward wave propagation, capturing similar wave the dynamics on naturally growing radially symmetric biofilms. Error bar shows the S.D, N= 5 measurements.

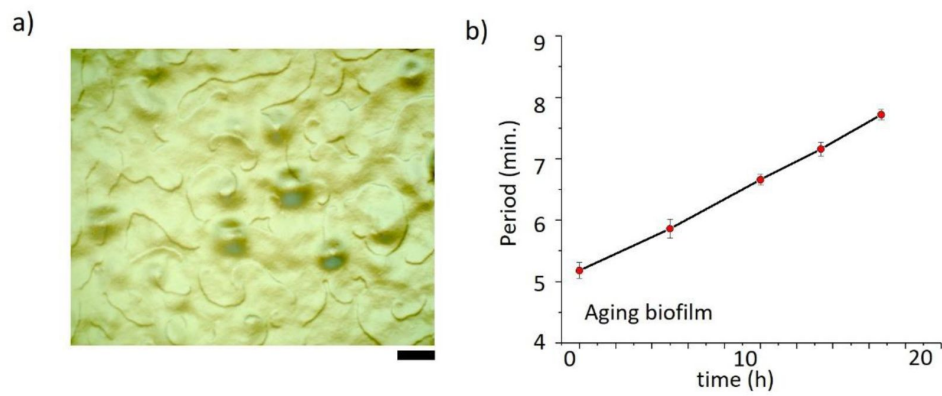


Figure 5

Age-Dependent Dynamics of Oscillations.

a) Sample image showing multiple pairs of spiral waves on a uniform biofilm surface. Scale bar: 100 μm . b) Period of the oscillations increases as the biofilm ages. Error bar shows the S.D. N= 10 measurements.

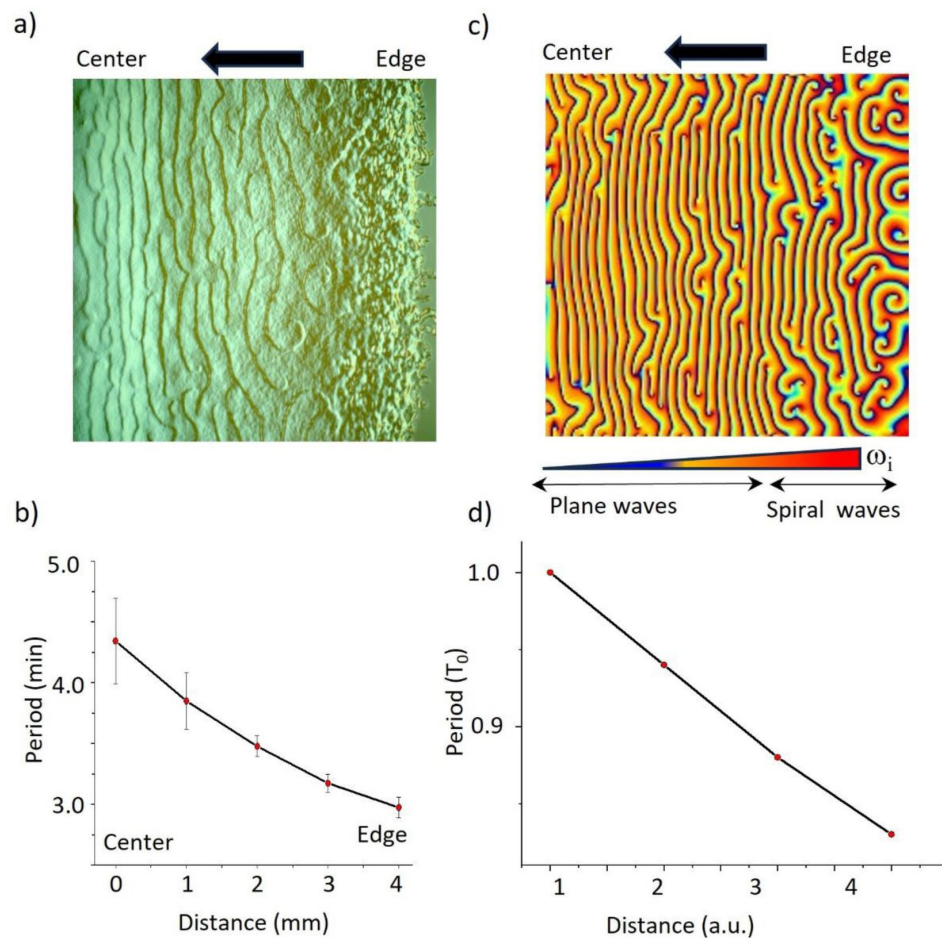


Figure 6.

Left-right symmetry breaking in naturally growing biofilms.

(a) Representative image of a bacterial biofilm at a late growth stage (3 days post-inoculation). Metachronal waves propagate towards the biofilm center, with the leading edge displaying chaotic dynamics. Following growth cessation, spiral waves gradually converge into planar waves propagating inward. (b) Oscillation period increases towards the biofilm center. Error bar indicates the S.D. $N=10$ measurements (c-d) Numerical simulation demonstrating the formation of planar waves and the left-right symmetry-breaking process, driven by a spatially varying intrinsic oscillation frequency(d), capturing dynamics observed in naturally growing biofilms.

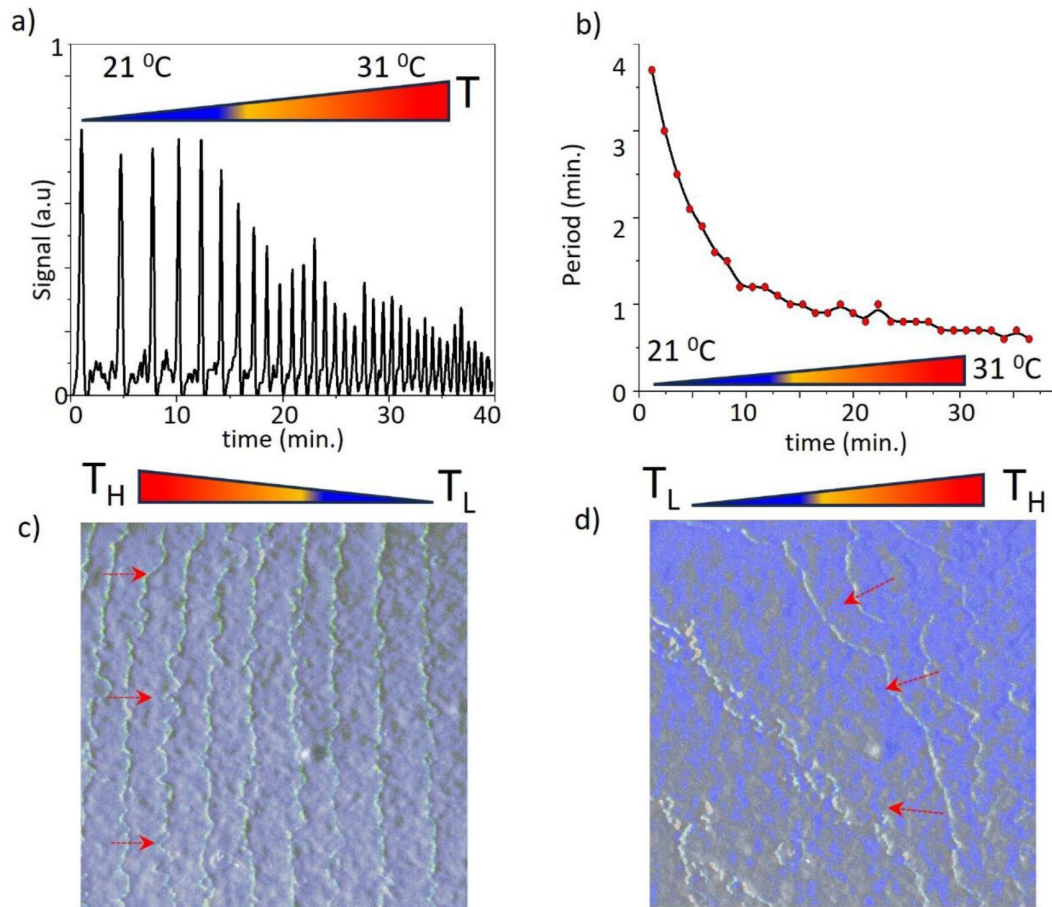


Figure 7

Temperature-Controlled Dynamics of Metachronal Waves and symmetry breaking.

a) Optical pulses from spiral waves on the biofilm surface as temperature increases from 21°C (T_L) to 31°C (T_H). b) An increase in temperature raises the oscillation frequency c, d) Controlling the propagation of the waves by creating a temperature gradient. Waves propagate from the warmer area (fast oscillating, T_H) to the colder region (slow oscillating, T_L). d) Reversing the temperature gradient changes the direction of propagations of the waves.

Discussion

Synchronization is a ubiquitous phenomenon in biological systems. A classic hallmark signature of this dynamical process is global synchronization, where all components of the system tend to merge in phase, as observed in fireflies, clocks and metronomes. Hydrodynamics or intrinsic activities of the interactions, however, introduce nonreciprocity into these tightly coupled systems, fundamentally altering the synchronization process. This action reaction asymmetry drives the emergence of large-scale propagating waves, often emerge as spiral waves in two dimensions. Remarkably, diverse systems from proteins in cell membrane to social amoebae exhibit similar dynamics, highlighting the special importance of these waves in biological systems particularly due to their transport capabilities.

From a symmetry perspective, nonreciprocity is the key feature in understanding these phenomena. Recent advancements in active matter physics have bridged various seemingly unrelated fields around this concept. Particularly, non-Hermitian physics, including PT-symmetry, provides a beneficial macroscopic metric for understanding a system's response through the classical formalism borrowed from quantum physics and optics. Despite the diverse nature of the dynamic processes and the varying types of interactions—from predator-prey dynamics, specific chemical reactions, to elasticity of active metamaterials—the common feature across these systems is to maintain phase difference between the fundamental fields that support limit cycle oscillations enabling collective motility and transport capabilities to the interacting active system.

We have observed that bacterial biofilm surfaces also known as active carpet exhibit similar dynamics due to coupled and cyclic pili extension and retraction activities. Although these nanoscale activities make the biophysical mechanisms complex and demanding to elucidate, the fundamental characteristic of the process remains consistent: the cyclic motion of bacterial pili forms limit cycle oscillations in phase space to disrupt local or action-reaction symmetry among the interacting components.

Using a phenomenological model, we have concentrated on identifying critical parameters that control the system dynamics and synchronization. We have also developed methods to manipulate and control the collective dynamics and propagation of metachronal waves by adjusting these critical parameters. From the perspective of active matter, the controllability of densely coupled active mechanical components is crucial, especially at low Reynolds numbers where controlling turbulence and managing long-range transport is vital. Biological systems can effectively address these challenges and precisely control active and programable transport⁶⁶. A deeper understanding of the underlying principles and critical symmetries governing this controllability could provide new insights into the complexities of biological systems. We speculate that these waves could promote active diffusion of oxygen and DNA. Further biophysical investigations are needed to delve deeper into these fascinating collective behaviors and its biological significance. Furthermore, the dynamical similarities between non-Hermitian physics and PT symmetry and the metachronal waves due to nonreciprocity are remarkable. Finally, we also hypothesize that the concepts developed in non-Hermitian physics could bring new perspective to dissect the complexities of these collective behaviors. Finally, our findings also raise several questions, particularly regarding the biological significance of these waves in the physiology of the biofilm. Rhythmic activities^{67,68} play a critical role in bacterial communities. We hypothesize that this synchronization may facilitate rhythmic behavior across the biofilm or enable expression and the transport of specific biomolecules under stressful conditions. Further studies are needed to elucidate the mechanisms underlying this emerging phenomenon.

Materials and Methods

Analysis of Pilin Proteins and Accessory Genes

Genetic analyses were conducted using the complete genome sequence of *Pseudomonas nitroreducens* strain L4 chromosome⁶⁹ (NCBI Reference Sequence: NZ_CP120376.1). The reference genomic region between pilB and tRNA-thr was specifically compared.

Imaging System

Time-lapse imaging was performed using multiple microscopy techniques, selected for their capability to visualize surface topography effectively. We observed that oblique and polarized illumination techniques significantly enhanced the visibility of surface waves. During the evaporation process, the roughness of the biofilm significantly varies. During the evaporation process, the roughness of the biofilm varies significantly. We observed that oblique illumination is more effective on rough surfaces, whereas polarized imaging provides clearer results on flat surfaces. Oblique imaging was primarily conducted using Zeiss Smartzoom digital microscopes and Nikon Stereo SMZ18 microscopes equipped with contrast illumination. Additional polarized imaging utilized a Zeiss Axio Imager M2m. Successive images were captured at intervals of 10 seconds. Fluorescence imaging for GFP was conducted using 488 nm blue-light excitation with GFP emission filters, and DNA staining was visualized using RFP emission filters and imaged with an Andor EMCCD camera. The typical optical magnification was 10X.

Image Processing

To enhance wave visibility, successive images taken at 10-second intervals were subtracted, and the resulting false-colored difference images were superimposed onto the original ones. The time-dependent responses of surface pulses were analyzed using ImageJ software.

Bacterial growth and Preparation

The *Pseudomonas nitroreducens* (PN) bacterial strain used in this study was initially isolated from environmental samples and identified through 16S rRNA sequencing. Verification was performed through comparison with the corresponding strain obtained from DSMZ. For optical imaging, PN bacteria were grown overnight in LB broth at 21 °C on a shaker, and subsequently, 100 µl drops of bacterial suspension were placed onto 10-cm, 1-day-old, 1.4% solid agar, NGM plates. To ensure slow evaporation of excess moisture, humidity was maintained at approximately 80%, and plate lids were partially left open. Nematode growth medium (NGM) plates were prepared following standard protocols.

Numerical Simulation

Numerical simulations based on the Kuramoto model were performed using custom-developed MATLAB code with finite difference algorithms. The original code for the active solid model was obtained from prior literature and modified for improved visualization. Through extensive parameter exploration, we determined that large lattice structures with fixed boundary conditions effectively reproduced spiral and planar wave dynamics.

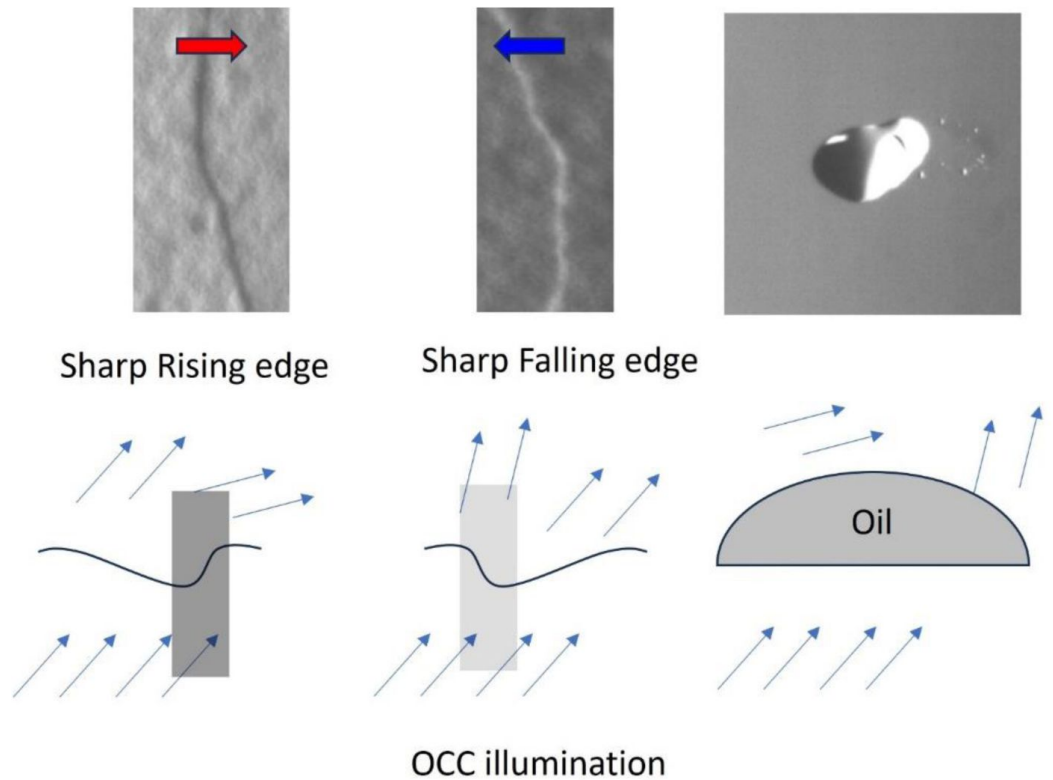
Data availability

The critical experimental data generated or analyzed during this study are provided as supporting video files. We did not generate additional data sets.

Software availability

All the codes used in the study will be available online.

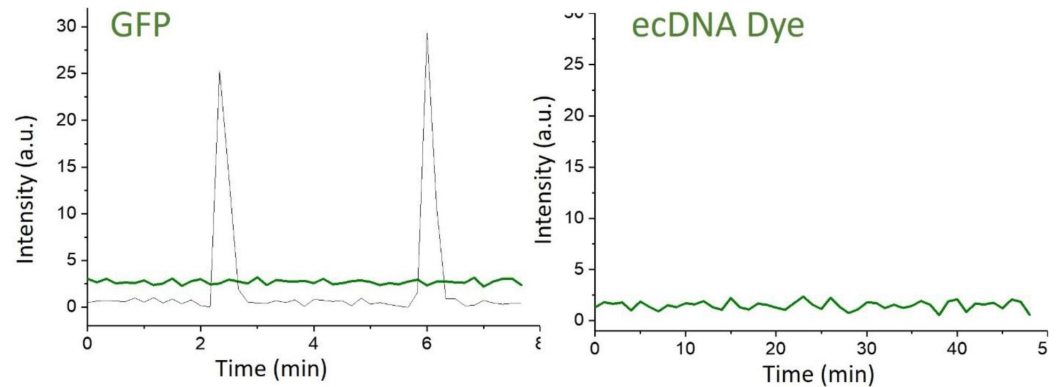
Supplementary Figures and Tables



Supplementary Figure 1.

Asymmetric surface deformation drives directional propagation.

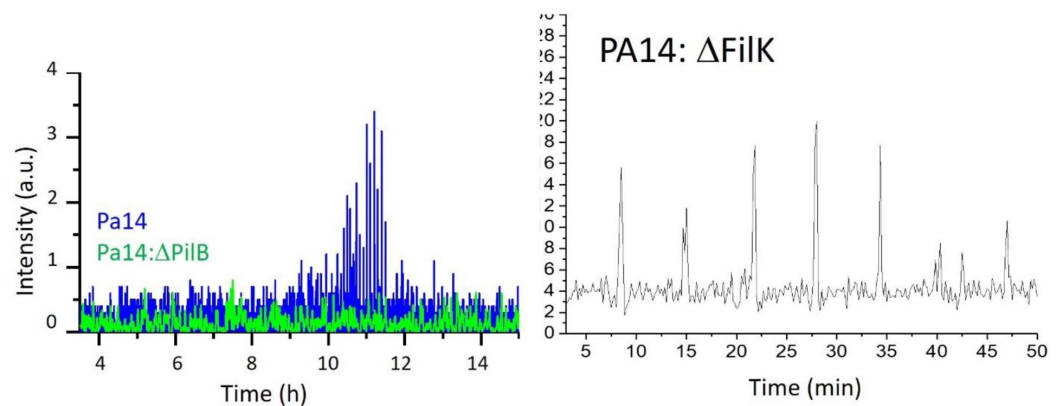
OCC imaging provides clear dark and bright optical contrasts, originating from oblique illumination and asymmetric scattering by surface waves. An oil droplet was used to investigate the shape-dependent propagation mechanism. Under oblique illumination, sharp rising edges generate dark contrasts, while falling edges create bright contrasts. Sharp edges correspond to rapid pili retraction, followed by gradual recovery.



Supplementary Figure 2.

GFP labeling and extracellular DNA staining were used to verify the origin of the optical scattering, specifically to distinguish between surface versus bulk oscillations in the biofilm.

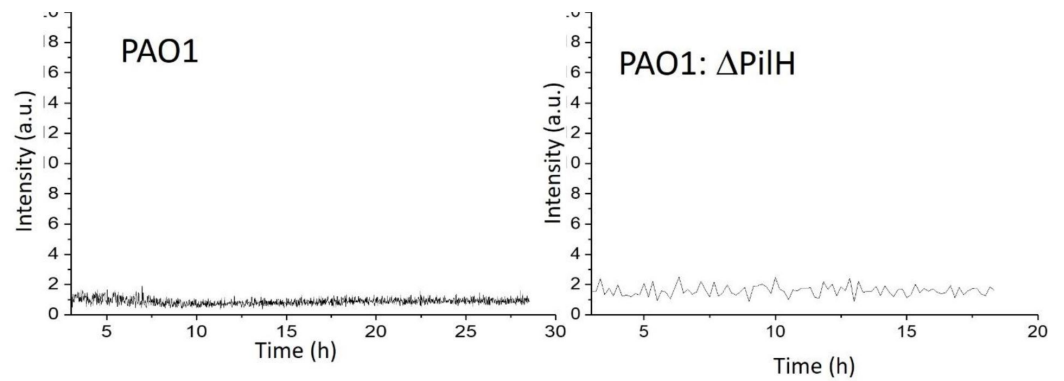
Neither fluorescence imaging method revealed oscillatory signals. Simultaneously recorded gray signals, indicating the presence of original waves on the biofilm, were superimposed for comparison.



Supplementary Figure 3.

In PA14 strains, mutations in pilB and fliK were introduced to identify the processes responsible for driving the oscillations.

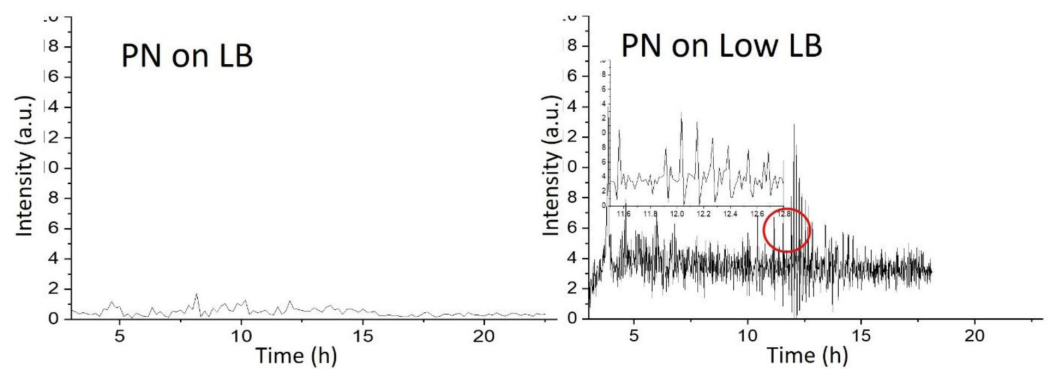
The pilB mutation abolished wave formation, whereas wave emergence persisted in the fliK mutant, which lacks functional flagella.



Supplementary Figure 4.

The PAO1 strain did not generate surface waves.

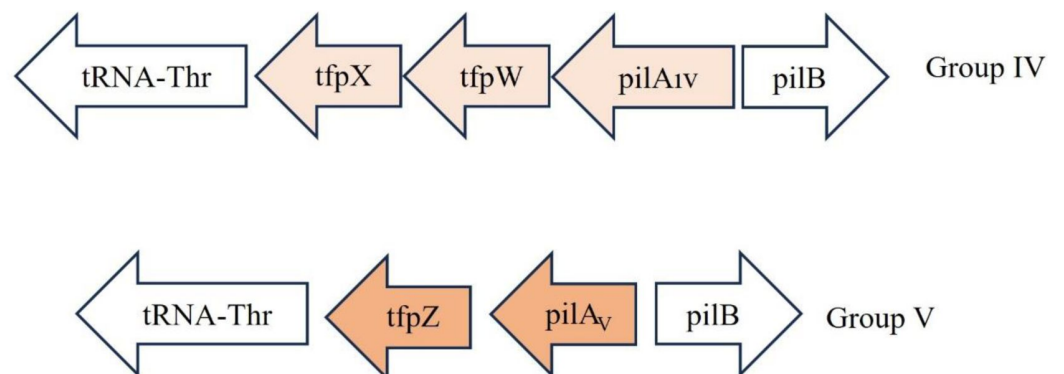
Additionally, the hyperpilated pilH mutant also failed to exhibit oscillatory wave behavior.



Supplementary Figure 5.

Regular LB plates do not provide sufficient conditions for wave generation in PN.

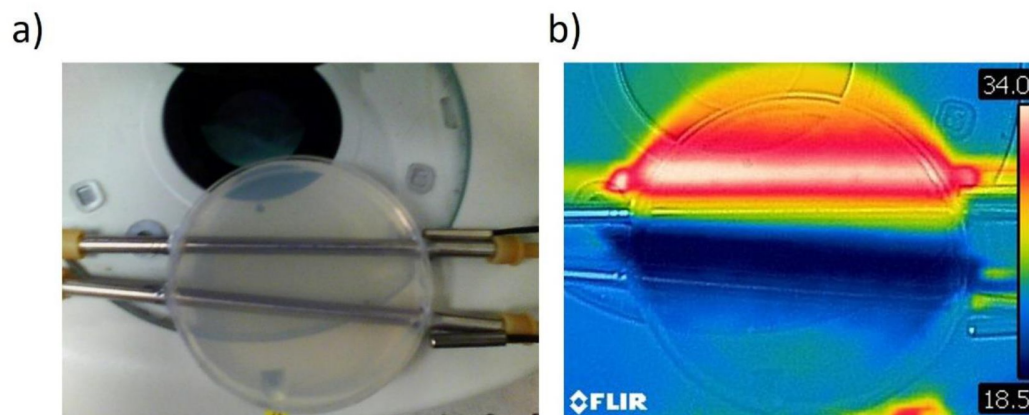
However, eliminating yeast extract and replacing tryptone with a low concentration (0.2 $\times$) peptone restored wave formation.



Supplementary Figure 6.

Comparative analysis of the genomic region encoding pili subunit groups reveals that PA14 contains Group 3 pili, while PN exhibits a Group 4 and 5-like structure.

Both strains also encode accessory proteins associated with unknown pilus function.



Supplementary Figure 7.

Optical and thermal images of an NGM plate equipped with metallic pipes.

A closed-loop water pumping-heating system was used to control the local temperature and create a temperature gradient between the pipes. The spacing between the pipes was varied to identify the optimal gradient profile.

Strain	Parent	Operation	Genotype
BAK132	PN	Isolated from environmental samples, transformed pUCP18-MCSgfpmut3	<i>ampR</i>
BAK133	PA14	Received from F. M. Ausubel lab	
BAK134	LD2222	Received from Lars Dietrich Lab ⁷⁰	PA14 Δ pilB
BAK135	LD2221	Received from Lars Dietrich Lab ⁷⁰	PA14 Δ flgK
BAK136	PAO1 PilH	Received from Joanne Engel	Δ PilH
BAK	PAO1	Received from DSMZ 10145GFP	

Table 1

List of strains used in this study.

Acknowledgements

This work was supported by Airforce Office of Scientific Research (AFOSR) under Grant No: FA9550-22-1-0431 with Program Manager Ali Sayir and Tübitak Project No: 121C159. S.K.O acknowledges the support from Air AFOSR Multidisciplinary University Research Initiative (MURI) Award No. FA9550-21-1-0202. We thank Zeiss, Turkey division and Kenan Doğru for digital microscopy installation and training.

Additional information

Author contributions

B.A. and A.Ko. initiated the project and performed the experiments. B.A., Y.I.Y., and M.B. Y.G. developed and executed numerical modeling. B.K. conducted 16S DNA sequencing and GFP labeling of PN. N.G. performed genome sequence analysis. E.T.G. and A.Ü. developed and conducted imaging for the temperature gradient control system. S.K.Ö. and C.K. conceptualized non-Hermitian and PT-symmetry principles. A.Ko. and B.A. drafted the manuscript, with contributions from all authors toward the final manuscript.

Additional files

Video 1. Time-lapse imaging of active *Pseudomonas nitroreducens* biofilm surfaces generating metachronal waves. Global left-right symmetry is broken, and merging spiral waves at the edge drive unidirectional planar waves toward the biofilm center. The wave wavelength is approximately 100 μm . Associated with Figure 1a. [🔗](#)

Video 2. Time-lapse imaging of circularly symmetric active biofilm surfaces generating inward-propagating metachronal waves. Associated with Figure 1b. [🔗](#)

Video 3. High magnification (100X) time-lapse imaging of biofilm surfaces illustrating bacterial displacement and optical contrast changes. [🔗](#)

Video 4. Time-lapse imaging of biofilm growth initiated from a single bacterium. After growth cessation, dense regions begin firing first, and waves dominate the biofilm surface. [🔗](#)

Video 5. Numerical simulation using the nonreciprocal Kuramoto model. Colors indicate oscillation phases. Spiral waves merge to form target waves, eventually converging into planar waves. Associated with Figure 2c. [🔗](#)

Video 6. Numerical simulation based on the active solid model. Emergence of spiral and planar waves within the lattice. Associated with Figure 2e. [🔗](#)

Video 7. Time-lapse imaging of controlling the transition between waves to target and spiral by manipulating the moisture of biofilm surface. Associated with Figure 3d-g. [🔗](#)

Video 8. Time-lapse imaging of a bacterial biofilm with broken left-right symmetry. Associated with Figure 6. [🔗](#)

Video 9. Time-lapse imaging of controlling propagating planar waves under a varying temperature gradient. Associated with Figure 7. [🔗](#)

Video 10. Numerical simulation of propagating planar waves influenced by a frequency gradient. Associated with Figure 7c. [🔗](#)

References

- 1 Wadhwa N., Berg H. C. (2022) **Bacterial motility: machinery and mechanisms** *Nat Rev Microbiol* **20**:161–173 <https://doi.org/10.1038/s41579-021-00626-4> | Google Scholar
- 2 Lauga E. (2016) **Bacterial Hydrodynamics** *Annu Rev Fluid Mech* **48**:105–130 <https://doi.org/10.1146/annurev-fluid-122414-034606> | Google Scholar
- 3 Lopez D., Lauga E. (2014) **Dynamics of swimming bacteria at complex interfaces** *Phys Fluids* **26**:071902 <https://doi.org/10.1063/1.4887255> | Google Scholar
- 4 Kearns D. B. (2010) **A field guide to bacterial swarming motility** *Nat Rev Microbiol* **8**:634–644 <https://doi.org/10.1038/nrmicro2405> | Google Scholar
- 5 Purcell E. M. (1977) **Life at low Reynolds number** *American Journal of Physics* **45**:3–11 <https://doi.org/10.1119/1.10903> | Google Scholar
- 6 Byron M. L., et al. (2021) **Metachronal Motion across Scales: Current Challenges and Future Directions** *Integr Comp Biol* **61**:1674–1688 <https://doi.org/10.1093/icb/icab105> | Google Scholar
- 7 Leoni M., Liverpool T. B. (2012) **Hydrodynamic synchronization of nonlinear oscillators at low Reynolds number** *Phys Rev E* **85**:040901 <https://doi.org/10.1103/PhysRevE.85.040901> | Google Scholar
- 8 Golestanian R., Yeomans J. M., Uchida N. (2011) **Hydrodynamic synchronization at low Reynolds number** *Soft Matter* **7**:3074–3082 <https://doi.org/10.1039/c0sm01121e> | Google Scholar
- 9 Sleigh M. A., Blake J. R., Liron N. (1988) **The Propulsion of Mucus by Cilia** *Am Rev Respir Dis* **137**:726–741 <https://doi.org/10.1164/ajrccm/137.3.726> | Google Scholar
- 10 Hirokawa N., Tanaka Y., Okada Y., Takeda S. (2006) **Nodal flow and the generation of left-right asymmetry** *Cell* **125**:33–45 <https://doi.org/10.1016/j.cell.2006.03.002> | Google Scholar
- 11 Cohen A. H., Holmes P. J., Rand R. H. (1982) **The Nature of the Coupling between Segmental Oscillators of the Lamprey Spinal Generator for Locomotion - a Mathematical-Model** *J Math Biol* **13**:345–369 <https://doi.org/10.1007/Bf00276069> | Google Scholar
- 12 Yaman Y. I., Ramanathan S. (2023) **Controlling human organoid symmetry breaking reveals signaling gradients drive segmentation clock waves** *Cell* **186**:513 <https://doi.org/10.1016/j.cell.2022.12.042> | Google Scholar
- 13 Wu Q., et al. (2023) **Active metamaterials for realizing odd mass density** *P Natl Acad Sci USA* **120**:e2209829120 <https://doi.org/10.1073/pnas.2209829120> | Google Scholar

- 14 Bender C. M., Boettcher S. (1998) **Real spectra in non-Hermitian Hamiltonians having PT symmetry** *Physical Review Letters* **80**:5243–5246 <https://doi.org/10.1103/PhysRevLett.80.5243> | [Google Scholar](#)
- 15 Chen Y. Y., Li X. P., Scheibner C., Vitelli V., Huang G. L. (2021) **Realization of active metamaterials with odd micropolar elasticity** *Nature Communications* **12**:5935 <https://doi.org/10.1038/s41467-021-26034-z> | [Google Scholar](#)
- 16 Fruchart M., Scheibner C., Vitelli V. (2023) **Odd Viscosity and Odd Elasticity** *Annual Review of Condensed Matter Physics* **14**:471–510 <https://doi.org/10.1146/annurev-conmatphys-040821-125506> | [Google Scholar](#)
- 17 Hickey D. J., Golestanian R., Vilfan A. (2023) **Nonreciprocal interactions give rise to fast cilium synchronization in finite systems** *P Natl Acad Sci USA* **120**:e2307279120 <https://doi.org/10.1073/pnas.2307279120> | [Google Scholar](#)
- 18 Brauns F., Marchetti M. C. (2024) **Nonreciprocal Pattern Formation of Conserved Fields** *Phys. Rev. X* **14**:021014 <https://doi.org/10.1103/PhysRevX.14.021014> | [Google Scholar](#)
- 19 Fruchart M., Hanai R., Littlewood P. B., Vitelli V. (2021) **Non-reciprocal phase transitions** *Nature* **592** <https://doi.org/10.1038/s41586-021-03375-9> | [Google Scholar](#)
- 20 Dinelli A., et al. (2023) **Non-reciprocity across scales in active mixtures** *Nat Commun* **14**:7035 <https://doi.org/10.1038/s41467-023-42713-5> | [Google Scholar](#)
- 21 You Z. H., Baskaran A., Marchetti M. C. (2020) **Nonreciprocity as a generic route to traveling states** *P Natl Acad Sci USA* **117**:19767–19772 <https://doi.org/10.1073/pnas.2010318117> | [Google Scholar](#)
- 22 Tang E., Agudo-Canalejo J., Golestanian R. (2021) **Topology Protects Chiral Edge Currents in Stochastic Systems** *Phys Rev X* **11**:031015 <https://doi.org/10.1103/PhysRevX.11.031015> | [Google Scholar](#)
- 23 Uchida N., Golestanian R. (2010) **Synchronization and Collective Dynamics in a Carpet of Microfluidic Rotors** *Physical Review Letters* **104**:178103 <https://doi.org/10.1103/PhysRevLett.104.178103> | [Google Scholar](#)
- 24 Hatano N., Nelson D. R. (1996) **Localization transitions in non-Hermitian quantum mechanics** *Physical Review Letters* **77**:570–573 <https://doi.org/10.1103/PhysRevLett.77.570> | [Google Scholar](#)
- 25 Ozdemir S. K., Rotter S., Nori F., Yang L. (2019) **Parity-time symmetry and exceptional points in photonics** *Nat Mater* **18**:783–798 <https://doi.org/10.1038/s41563-019-0304-9> | [Google Scholar](#)
- 26 El-Ganainy R., et al. (2018) **Non-Hermitian physics and PT symmetry** *Nat Phys* **14**:11–19 <https://doi.org/10.1038/Nphys4323> | [Google Scholar](#)
- 27 Miri M. A., Alù A. (2019) **Exceptional points in optics and photonics** *Science* **363**:42 <https://doi.org/10.1126/science.aar7709> | [Google Scholar](#)
- 28 Naghiloo M., Abbasi M., Joglekar Y. N., Murch K. W. (2019) **Quantum state tomography across the exceptional point in a single dissipative qubit** *Nat Phys* **15**:1232 <https://doi.org/10.1038/s41567-019-0652-z> | [Google Scholar](#)

- 29 Zhang J. W., et al. (2022) **Dynamical control of quantum heat engines using exceptional points** *Nature Communications* **13**:6225 <https://doi.org/10.1038/s41467-022-33667-1> | Google Scholar
- 30 Ding L. Y., et al. (2021) **Experimental Determination of -Symmetric Exceptional Points in a Single Trapped Ion** *Physical Review Letters* **126**:083604 <https://doi.org/10.1103/PhysRevLett.126.083604> | Google Scholar
- 31 Wu Y., et al. (2019) **Observation of parity-time symmetry breaking in a single-spin system** *Science* **364**:878 <https://doi.org/10.1126/science.aaw8205> | Google Scholar
- 32 Ergoktas M. S., et al. (2022) **Topological engineering of terahertz light using electrically tunable exceptional point singularities** *Science* **376**:184 <https://doi.org/10.1126/science.abn6528> | Google Scholar
- 33 Soleymani S., et al. (2022) **Chiral and degenerate perfect absorption on exceptional surfaces** *Nature Communications* **13**:599 <https://doi.org/10.1038/s41467-022-27990-w> | Google Scholar
- 34 Zhong Q., et al. (2019) **Sensing with Exceptional Surfaces in Order to Combine Sensitivity with Robustness** *Physical Review Letters* **122**:153902 <https://doi.org/10.1103/PhysRevLett.122.153902> | Google Scholar
- 35 Lau H. K., Clerk A. A. (2018) **Fundamental limits and non-reciprocal approaches in non-Hermitian quantum sensing** *Nature Communications* **9**:4320 <https://doi.org/10.1038/s41467-018-06477-7> | Google Scholar
- 36 Tlusty T. (2021) **Exceptional topology in ordinary soft matter** *Phys Rev E* **104**:025002 <https://doi.org/10.1103/PhysRevE.104.025002> | Google Scholar
- 37 Biktashev V. N., Brindley J., Holden A. V., Tsyganov M. A. (2004) **Pursuit-evasion predator-prey waves in two spatial dimensions** *Chaos* **14**:988–994 <https://doi.org/10.1063/1.1793751> | Google Scholar
- 38 Loiseau E., et al. (2020) **Active mucus-cilia hydrodynamic coupling drives self-organization of human bronchial epithelium** *Nat Phys* **16**:1158 <https://doi.org/10.1038/s41567-020-0980-z> | Google Scholar
- 39 Guzmán-Lastra F., Löwen H., Mathijssen A. J. T. M. (2021) **Active carpets drive non-equilibrium diffusion and enhanced molecular fluxes** *Nature Communications* **12**:1906 <https://doi.org/10.1038/s41467-021-22029-y> | Google Scholar
- 40 Kim M. J., Breuer K. S. (2007) **Use of bacterial carpets to enhance mixing in microfluidic systems** *J Fluid Eng-T Asme* **129**:319–324 <https://doi.org/10.1115/1.2427083> | Google Scholar
- 41 Tan T. H., et al. (2022) **Odd dynamics of living chiral crystals** *Nature* **607**:287 <https://doi.org/10.1038/s41586-022-04889-6> | Google Scholar
- 42 Gregor T., Fujimoto K., Masaki N., Sawai S. (2010) **The Onset of Collective Behavior in Social Amoebae** *Science* **328**:1021–1025 <https://doi.org/10.1126/science.1183415> | Google Scholar
- 43 Dreher A., Aranson I. S., Kruse K. (2014) **Spiral actin-polymerization waves can generate amoeboidal cell crawling** *New J Phys* **16**:055007 <https://doi.org/10.1088/1367-2630/16/5/055007> | Google Scholar

- 44 Ford H. Z., Manhart A., Chubb J. R. (2023) **Controlling periodic long-range signalling to drive a morphogenetic transition** *eLife* **12**:doi:ARTN e83796 <https://doi.org/10.7554/eLife.83796> | [Google Scholar](#)
- 45 Bull M. S., Kroo L. A., Prakash M. (2021) **Excitable mechanics embodied in a walking cilium** *arXiv* :2107.02930 [Google Scholar](#)
- 46 Kühn M. J., et al. (2021) **Mechanotaxis directs aeruginosa twitching motility** *P Natl Acad Sci USA* **118**:e2101759118 <https://doi.org/10.1073/pnas.2101759118> | [Google Scholar](#)
- 47 Kühn M. J., et al. (2023) **Two antagonistic response regulators control polarization during mechanotaxis** *Embo J* **42** <https://doi.org/10.15252/emboj.2022112165> | [Google Scholar](#)
- 48 Xu H. R., Huang Y. L., Zhang R., Wu Y. L. (2023) **Autonomous waves and global motion modes in living active solids** *Nat Phys* **19**:46 <https://doi.org/10.1038/s41567-022-01836-0> | [Google Scholar](#)
- 49 Liu S. Q., Li Y., Wang Y. H., Wu Y. L. (2024) **Emergence of large-scale mechanical spiral waves in bacterial living matter** *Nat Phys* **20** <https://doi.org/10.1038/s41567-024-02457-5> | [Google Scholar](#)
- 50 Rana N., Golestanian R. (2024) **Defect Solutions of the Nonreciprocal Cahn-Hilliard Model: Spirals and Targets** *Physical Review Letters* **133**:078301 <https://doi.org/10.1103/PhysRevLett.133.078301> | [Google Scholar](#)
- 51 Demir E., Yaman Y. I., Basaran M., Kocabas A. (2020) **Dynamics of pattern formation and emergence of swarming in *Caenorhabditis elegans*** *eLife* **9**:doi:ARTN e52781 <https://doi.org/10.7554/eLife.52781> | [Google Scholar](#)
- 52 Kus J. V., Tullis E., Cvitkovitch D. G., Burrows L. L. (2004) **Significant differences in type IV pilin allele distribution among *Pseudomonas aeruginosa* isolates from cystic fibrosis (CF) versus non-CF patients** *Microbiology* **150**:1315–1326 <https://doi.org/10.1099/mic.0.26822-0> | [Google Scholar](#)
- 53 Meron E. (1992) **Pattern-Formation in Excitable Media** *Phys Rep* **218**:1–66 [https://doi.org/10.1016/0370-1573\(92\)90098-K](https://doi.org/10.1016/0370-1573(92)90098-K) | [Google Scholar](#)
- 54 Tan T. H., et al. (2020) **Topological turbulence in the membrane of a living cell** *Nat Phys* **16**:657–662 <https://doi.org/10.1038/s41567-020-0841-9> | [Google Scholar](#)
- 55 Aranson I. S., Rabinovich M. I. (1990) **Dynamics of spiral waves in non-equilibrium media** *Journal of Physics A: Mathematical and General* **23**:299 <https://doi.org/10.1088/0305-4470/23/3/014> | [Google Scholar](#)
- 56 Paultet J. E., Ermentrout G. B. (1994) **Stable Rotating Waves in Two-Dimensional Discrete Active Media** *SIAM Journal on Applied Mathematics* **54**:1720–1744 <https://doi.org/10.1137/S0036139993250683> | [Google Scholar](#)
- 57 Chakrabarti B., Furthauer S., Shelley M. J. (2022) **A multiscale biophysical model gives quantized metachronal waves in a lattice of beating cilia** *P Natl Acad Sci USA* **119**:e2113539119 <https://doi.org/10.1073/pnas.2113539119> | [Google Scholar](#)
- 58 Baconnier P., et al. (2022) **Selective and collective actuation in active solids** *Nat Phys* **18**:1234 <https://doi.org/10.1038/s41567-022-01704-x> | [Google Scholar](#)

- 59 Kim P. J., Ko T. W., Jeong H., Moon H. T. (2004) **Pattern formation in a two-dimensional array of oscillators with phase-shifted coupling** *Phys Rev E* **70**:065201 <https://doi.org/10.1103/PhysRevE.70.065201> | Google Scholar
- 60 Gupta R. K., Kant R., Soni H., Sood A. K., Ramaswamy S. (2022) **Active nonreciprocal attraction between motile particles in an elastic medium** *Phys Rev E* **105**:064602 <https://doi.org/10.1103/PhysRevE.105.064602> | Google Scholar
- 61 Kim S., Park S. H., Ryu C. S. (1997) **Multistability in coupled oscillator systems with time delay** *Physical Review Letters* **79**:2911–2914 <https://doi.org/10.1103/PhysRevLett.79.2911> | Google Scholar
- 62 Ohta H., Sasa S. (2008) **Critical phenomena in globally coupled excitable elements** *Phys Rev E* **78**:065101 <https://doi.org/10.1103/PhysRevE.78.065101> | Google Scholar
- 63 Gu F. C., Guiselin B., Bain N., Zuriguel I., Bartolo D. (2025) **Emergence of collective oscillations in massive human crowds** *Nature* **638** <https://doi.org/10.1038/s41586-024-08514-6> | Google Scholar
- 64 Rana N., Golestanian R. (2024) **Defect interactions in the non-reciprocal Cahn-Hilliard model** *New J Phys* **26**:123008 <https://doi.org/10.1088/1367-2630/ad9859> | Google Scholar
- 65 Wang G. Q., et al. (2010) **Target-wave to spiral-wave pattern transition in a discrete Belousov-Zhabotinsky reaction driven by inactive resin beads** *Phys Rev E* **82**:045201 <https://doi.org/10.1103/PhysRevE.82.045201> | Google Scholar
- 66 Yang F., Liu S. C., Lee H. J., Phillips R., Thomson M. (2025) **Dynamic flow control through active matter programming language** *Nat Mater* **24** <https://doi.org/10.1038/s41563-024-02090-w> | Google Scholar
- 67 Sartor F., Kovács A. T. (2022) **Rhythmic Spatial Self-Organization of Bacterial Colonies** *Mbio* **13** <https://doi.org/10.1128/mbio.01703-22> | Google Scholar
- 68 Eelderink-Chen Z., et al. (2021) **A circadian clock in a nonphotosynthetic prokaryote** *Sci Adv* **7**:eabe2086 <https://doi.org/10.1126/sciadv.abe2086> | Google Scholar
- 69 Liu H. Y., Zhuang L. B., Zeng Q. C. (2024) **Complete genome sequence data of L4, an endophyte isolated from cotton plants** *Data Brief* **55**:110639 <https://doi.org/10.1016/j.dib.2024.110639> | Google Scholar
- 70 Hölscher T., et al. (2015) **Motility, Chemotaxis and Aerotaxis Contribute to Competitiveness during Bacterial Pellicle Biofilm Development** *J Mol Biol* **427**:3695–3708 <https://doi.org/10.1016/j.jmb.2015.06.014> | Google Scholar

Author information

Baha Altın

Department of Physics, Koç University, Istanbul, Türkiye

Yusuf Ilker Yaman

Department of Physics, Koç University, Istanbul, Türkiye, Sciences and Engineering Program,
Harvard University, Cambridge, United States

Enes Talha Günay

Department of Physics, Koç University, Istanbul, Türkiye

Alp Ünlü

Department of Physics, Koç University, Istanbul, Türkiye

Yiğithan Gediz

Department of Physics, Koç University, Istanbul, Türkiye

Neslihan Gedik

Department of Physics, Undergraduate Program, Boğaziçi University, Istanbul, Türkiye

Bora Karadaş

Department of Physics, Koç University, Istanbul, Türkiye

Mustafa Başaran

Department of Physics, Koç University, Istanbul, Türkiye, Sciences and Engineering Program,
Harvard University, Cambridge, United States

Coşkun Kocabaş

Department of Materials, University of Manchester, Manchester, United Kingdom

Şahin Özdemir

Department of Engineering Science and Mechanics, The Pennsylvania State University,
University Park, United States, Department of Electrical and Computer Engineering, Saint
Louis University, St Louis, United States

Aşkın Kocabaş

Department of Physics, Koç University, Istanbul, Türkiye, Koç University Surface Science and
Technology Center, Koç University, Istanbul, Türkiye

For correspondence: akocabas@ku.edu.tr

Editors

Reviewing Editor

Shashi Thutupalli

Simons Centre for the Study of Living Machines, National Centre for Biological Sciences (TIFR),
Bangalore, India

Senior Editor

Felix Campelo

Universitat Pompeu Fabra, Barcelona, Spain

Reviewer #1 (Public review):

Summary:

Overall, this is an interesting paper. The authors have found multiple experimental knobs to perturb a mechanical wave behavior driven by pilli feedback. The authors framed this as nonreciprocal interactions - while I can see how nonreciprocity could play a role - what about mechanical feedback? Phenomenological models are fine, but a lack of mechanistic understanding is a weakness. I think it will be more interesting to frame the model based on potential mechanochemical feedback to understand microscopic mechanisms. Regardless, more can be done to better constrain the model through finding knobs to explain experimental observations (in Figures 3, 4, 5, and 7).

Strengths:

The report of mechanical waves in bacterial collectives. The mechanism has potential application in a multicellular context, such as morphogenesis.

Weaknesses:

My most serious concern is about left-right symmetry breaking. I fail to see how the data in Figure 6 shows LR symmetry breaking. All they show is in-out directionality, which is a boundary condition. LR SM means breaking of mirror symmetry - the pattern cannot be superimposed on its mirror image using only rigid body transformations (translation and rotation) - as far as I am aware, this condition is not satisfied in this pattern-forming system.

<https://doi.org/10.7554/eLife.107609.1.sa2>

Reviewer #2 (Public review):

Summary:

This manuscript by Altin et al. examines the dynamics of bacterial assemblies, building on previously published work documenting mechanical spiral waves. The authors show that the emergent dynamics can be influenced by various factors, including the strain of bacteria and water content in the sample. While the topic of this paper would be of broad interest, and the preliminary results are certainly interesting, various aspects of this paper are underdeveloped and require further exploration.

Strengths:

One of the nice features of this system is the ability to transition between the different states based on the addition or withdrawal of water. The authors use a similar experimental model system and mathematical model to previously published work (Reference 49), but extend by showing that the behaviour can be modified through simple interventions. Specifically, the authors show that adding water droplets or drying the sample through heating can result in changes in the observed wave structure. This represents a possible way of controlling active matter.

The mathematical model proposed in this paper involves a phase-oscillator model of Kuramoto-style coupling (similar to previously reported models). A non-reciprocal phase lag is introduced in order to facilitate the patterns seen in experiments. The qualitative agreement in the behaviour is quite striking, showing both spiral waves and travelling waves.

Weaknesses:

The principal observation of the paper - that spiral waves emerge in these systems and can be controlled in various ways - is not linked to microscale dynamics at the cell level. It is recognised that hydrodynamics can introduce non-reciprocity, an essential ingredient of this model. However, in this work the authors have not identified a physical mechanism for the

lag, e.g., either through steric interactions or hydrodynamic disturbances. This is also relevant in the phase oscillator modelling section. In low Reynolds number flows, dynamics are instantaneously determined. In this light, what does the phase lag term represent? What is the origin of the coupling term, b ? Can this be varied systematically or derived from experimental measurements or parameters?

Classification of wave properties is an important aspect of this paper, but is not accomplished in a quantitative sense. What is the method for distinguishing between travelling and spiral waves? There is a range of quantitative tools that could be used to investigate these dynamics (and also compare quantitatively with the models). For example, examining the correlation functions and order parameters could assist with the extraction of wave features (see extensive literature on oscillator models).

The methodology of changing the dynamics through moisture content appears to be slightly underdeveloped, e.g., adding water involves a droplet, and removing water is accomplished by heating (which presumably could cause other effects). Could the dynamics not be controlled more directly by varying the humidity? At the same time, the authors also mention that temperature itself plays a role in shaping the behaviour. What is the mechanism for this? Is it just through evaporation? Since the frequency increases with temperature, could it just be that activity increases with temperature?

<https://doi.org/10.7554/eLife.107609.1.sa1>

Reviewer #3 (Public review):

Summary:

This manuscript presents a novel investigation into unidirectionally propagating waves observed on the surface of *Pseudomonas nitroreducens* bacterial biofilms. The authors explore how these waves, initially spiral in form, transition into combinations of spiral, target, and planar patterns. The study identifies the periodic extension-retraction cycles of type IV pili as the driving mechanism for wave propagation, which preferentially moves from the colony's edge to its center. Furthermore, the manuscript proposes two theoretical models—a phase-oscillator model and a continuum active solid model—to reproduce these phenomena, and demonstrates how external manipulations (e.g., water droplets, temperature, PEG) can control wave patterns and direction, often correlating with oscillation frequency gradients. The work aims to bridge the fields of active-matter physics and bacterial biophysics by providing both experimental observations and theoretical frameworks for understanding these complex biological wave phenomena.

Strengths:

The experimental discovery of unidirectionally propagating waves on bacterial biofilms is highly intriguing and represents a significant contribution to both microbiology and active-matter physics. The detailed observations of wave pattern transitions (spiral to target to planar) and their response to various environmental perturbations (water, temperature, PEG) provide valuable empirical data. The identification of type IV pili as the driving force offers a concrete biological mechanism. The observed correlation between frequency gradients and wave direction is a compelling finding with potential for broader implications in understanding biological pattern formation. This work has the potential to stimulate further research in the collective behavior of living systems and the physical principles underlying biological organization.

Weaknesses:

The manuscript attempts to link unidirectional wave propagation to non-reciprocal couplings but ultimately shows that the wave direction is determined by the gradient of the oscillation frequency. The couplings in the two theoretical models are both isotropic and thus cannot dictate the wave direction. A clear distinction should be made between non-reciprocity as a source of wave generation and non-uniformity as a controlling factor of wave direction.

The relationship between the phase oscillator model and the active solid model is unclear. Given that U and P are both dynamical variables evolving in three-dimensional space, defining the phase Φ precisely in the phase space spanned by U and P could be challenging. A graphical illustration of the definition of Φ would be beneficial. To ensure reproducibility of the numerical results, the parameter values used in the numerical simulations and an explicit definition of the elastic force in the active solid model should be provided.

The link between the theoretical models and experimental results is weak. For example, the propagation of the kink from the lower to the higher part of the surface (Figure 1e) could be addressed within the framework of the active solid model. The mechanism of transition from spiral to target waves (Figure 3a, b)) requires clarification, identifying which model parameter is crucial for inducing this transition. The wave propagation toward the lower frequency side is numerically demonstrated using the phase oscillator model, but a physical or intuitive explanation for this phenomenon is missing. Also, the wave transitions induced by the addition of water droplets and temperature rise are not linked to specific parameters in the theoretical models.

<https://doi.org/10.7554/eLife.107609.1.sa0>